

## TITU, Romania

WMO: 154190

Lat: 44.6528N

Lon: 25.5792E

Elev: 160

StdP: 99.42

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.6      | -9.8 | -14.3                         | 1.1 | -11.6 | -11.8 | 1.4 | -8.9 | 14.5                  | 0.9  | 10.2 | -1.2 | 1.2                   | 230  | 0.447 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.9                   | 33.4            | 21.4 | 32.1 | 21.1 | 30.6 | 20.6 | 23.1                | 30.8 | 22.2 | 29.7 | 21.6 | 28.7 | 1.6                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.7                            | 15.7 | 25.9 | 20.0 | 14.9 | 25.0 | 19.3 | 14.3 | 24.3 | 69.1          | 31.1 | 66.0 | 29.6 | 63.4 | 28.8 | 26.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 9.4           | 7.3  | 6.0 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |     | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.5        | -2.1  | 2.0   | 6.4   | 11.2  | 17.2  | 21.5  | 22.7  | 22.7  | 17.7  | 11.5  | 5.4   | 1.1   |  |  |
|                                                                        | DBStd       | 9.34        | 4.64  | 4.63  | 4.46  | 4.30  | 3.09  | 2.83  | 2.72  | 2.81  | 3.67  | 3.89  | 4.30  | 3.93  |  |  |
|                                                                        | HDD10.0     | 1218        | 375   | 226   | 125   | 37    | 1     | 0     | 0     | 0     | 0     | 29    | 149   | 276   |  |  |
|                                                                        | HDD18.3     | 2935        | 633   | 458   | 368   | 216   | 58    | 6     | 2     | 3     | 57    | 212   | 389   | 534   |  |  |
|                                                                        | CDD10.0     | 1763        | 0     | 1     | 15    | 72    | 223   | 346   | 394   | 394   | 231   | 77    | 9     | 0     |  |  |
|                                                                        | CDD18.3     | 440         | 0     | 0     | 0     | 1     | 22    | 102   | 138   | 139   | 37    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 4567        | 0     | 0     | 2     | 48    | 263   | 978   | 1386  | 1449  | 407   | 34    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1778        | 0     | 0     | 0     | 5     | 56    | 360   | 577   | 650   | 126   | 4     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.0         | 2.2   | 2.3   | 2.6   | 2.5   | 2.3   | 1.6   | 1.4   | 1.4   | 1.7   | 1.7   | 1.8   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 624         | 37    | 28    | 37    | 53    | 71    | 80    | 73    | 46    | 58    | 51    | 38    | 45    |  |  |
|                                                                        | PrecMax     | 961         | 75    | 80    | 79    | 149   | 171   | 194   | 187   | 193   | 121   | 104   | 100   | 144   |  |  |
|                                                                        | PrecMin     | 342         | 2     | 1     | 1     | 19    | 13    | 28    | 25    | 3     | 8     | 0     | 2     | 0     |  |  |
|                                                                        | PrecStd     | 145         | 21    | 19    | 25    | 34    | 40    | 37    | 43    | 37    | 36    | 30    | 26    | 33    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.6  | 17.4  | 22.7  | 27.1  | 30.7  | 34.1  | 35.0  | 35.8  | 32.4  | 27.0  | 20.3  | 13.8  |  |  |
|                                                                        |             | MCWB        | 7.1   | 11.6  | 13.5  | 16.5  | 18.8  | 22.6  | 22.4  | 22.3  | 19.9  | 17.1  | 14.4  | 9.3   |  |  |
|                                                                        | 2%          | DB          | 7.7   | 13.6  | 19.4  | 24.5  | 28.3  | 32.1  | 33.3  | 33.4  | 29.7  | 23.6  | 16.6  | 10.5  |  |  |
|                                                                        |             | MCWB        | 5.5   | 9.3   | 11.4  | 15.6  | 18.5  | 21.9  | 21.8  | 21.0  | 18.6  | 16.0  | 12.3  | 7.9   |  |  |
|                                                                        | 5%          | DB          | 5.3   | 10.8  | 16.6  | 21.7  | 26.4  | 30.5  | 31.6  | 32.2  | 27.6  | 21.0  | 13.5  | 8.1   |  |  |
|                                                                        |             | MCWB        | 4.3   | 7.5   | 10.1  | 13.9  | 17.6  | 21.2  | 21.3  | 20.8  | 18.2  | 14.7  | 10.9  | 6.2   |  |  |
|                                                                        | 10%         | DB          | 3.6   | 8.3   | 14.1  | 19.4  | 24.4  | 28.7  | 29.9  | 30.5  | 25.5  | 18.6  | 11.2  | 5.8   |  |  |
|                                                                        |             | MCWB        | 2.6   | 6.1   | 9.1   | 13.0  | 16.8  | 20.5  | 20.8  | 20.5  | 17.5  | 13.6  | 9.4   | 4.4   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.6   | 12.4  | 14.1  | 17.2  | 20.5  | 24.1  | 24.1  | 23.9  | 21.8  | 17.8  | 14.6  | 10.0  |  |  |
|                                                                        |             | MCDB        | 10.1  | 16.7  | 21.7  | 25.7  | 28.3  | 31.6  | 32.2  | 32.7  | 29.8  | 24.8  | 18.9  | 12.2  |  |  |
|                                                                        | 2%          | WB          | 5.8   | 9.9   | 12.1  | 16.0  | 19.5  | 22.7  | 23.0  | 22.7  | 19.7  | 16.5  | 13.0  | 8.4   |  |  |
|                                                                        |             | MCDB        | 7.1   | 13.3  | 18.4  | 24.1  | 26.5  | 30.3  | 30.7  | 30.5  | 27.1  | 22.6  | 15.8  | 10.6  |  |  |
|                                                                        | 5%          | WB          | 4.4   | 8.0   | 10.7  | 14.8  | 18.5  | 21.9  | 22.2  | 21.8  | 18.9  | 15.3  | 11.5  | 6.6   |  |  |
|                                                                        |             | MCDB        | 5.4   | 10.6  | 16.0  | 20.7  | 24.8  | 29.2  | 29.3  | 29.6  | 25.5  | 20.1  | 13.4  | 8.3   |  |  |
|                                                                        | 10%         | WB          | 2.9   | 6.4   | 9.3   | 13.5  | 17.5  | 21.2  | 21.5  | 21.1  | 18.1  | 14.0  | 9.8   | 4.9   |  |  |
|                                                                        |             | MCDB        | 3.5   | 8.2   | 13.5  | 18.2  | 23.2  | 27.9  | 28.1  | 28.8  | 24.8  | 17.6  | 11.4  | 6.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.5   | 8.7   | 10.5  | 11.3  | 11.7  | 12.1  | 12.9  | 13.6  | 12.5  | 11.2  | 7.8   | 6.8   |  |  |
|                                                                        |             | MCDBR       | 9.6   | 13.0  | 16.0  | 16.2  | 15.2  | 14.8  | 15.2  | 16.2  | 16.0  | 15.7  | 12.7  | 10.7  |  |  |
|                                                                        | 5% WB       | MCWBR       | 7.2   | 8.6   | 8.6   | 8.1   | 6.5   | 5.5   | 5.3   | 5.4   | 6.3   | 8.3   | 8.2   | 8.0   |  |  |
|                                                                        |             | MCWBR       | 8.8   | 12.7  | 14.8  | 14.9  | 13.6  | 14.1  | 13.7  | 14.5  | 14.1  | 14.7  | 11.3  | 11.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.336 | 0.368 | 0.403 | 0.451 | 0.428 | 0.436 | 0.446 | 0.462 | 0.422 | 0.393 | 0.367 | 0.340 |  |  |
|                                                                        |             | taud        | 2.299 | 2.219 | 2.179 | 2.070 | 2.212 | 2.226 | 2.207 | 2.153 | 2.231 | 2.291 | 2.311 | 2.317 |  |  |
|                                                                        | Ebn at Noon |             | 760   | 799   | 820   | 810   | 845   | 837   | 823   | 792   | 793   | 763   | 715   | 713   |  |  |
|                                                                        |             | Edh at Noon | 83    | 108   | 128   | 155   | 139   | 138   | 139   | 142   | 121   | 100   | 83    | 75    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.39  | 2.22  | 3.44  | 4.66  | 5.76  | 6.25  | 6.33  | 5.63  | 4.09  | 2.68  | 1.49  | 1.14  |  |  |
|                                                                        | RadStd      |             | 0.20  | 0.33  | 0.37  | 0.47  | 0.46  | 0.45  | 0.43  | 0.35  | 0.33  | 0.33  | 0.19  | 0.17  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (69 neighbors) | +0.49 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -142    | +63     | N/A     |

Nomenclature: See separate page